

Syllabus for

Academic year 2021/2022 >

TMA321 - Mathematical statistics ⚠

Matematisk statistik

Syllabus adopted 2021-01-15 by Head of Programme (or corresponding)

Owner: [TKTFY](#)

4,5 Credits

Grading: TH - Pass with distinction (5), Pass with credit (4), Pass (3), Fail

Education cycle: First-cycle

Main field of study: Mathematics, Engineering Physics

Department: 11 - MATHEMATICAL SCIENCES

Teaching language: Swedish

Application code: 57137

Open for exchange students: No

Maximum participants: 130

Only students with the course round in the programme plan

Credit distribution

Module	Sp1	Sp2	Sp3	Sp4	Summer course ⓘ	No Sp	Examination dates ⓘ
0194 Examination 4,5 c Grading: TH				4,5 c			Wed 01/06-2022 am J, Sat 09/10-2021 pm J ⚠ Mon 15/08-2022 pm J ⚠

In programs

[TKTFY ENGINEERING PHYSICS, Year 1 \(compulsory\)](#)

[TKTFY ENGINEERING PHYSICS, Year 2 \(compulsory\)](#)

Examiner:

Johan Jonasson

 [Go to course homepage](#)

Eligibility

General entry requirements for bachelor's level (first cycle)

Applicants enrolled in a programme at Chalmers where the course is included in the study programme are exempted from fulfilling the requirements above.

Specific entry requirements

The same as for the programme that owns the course.

Applicants enrolled in a programme at Chalmers where the course is included in the study programme are exempted from fulfilling the requirements above.

Course specific prerequisites

Basic courses in one variable and multivariable analysis, and linear algebra.

Aim

Our aim is to give a new view of measurement and information focusing on the variation and uncertainty and on efficient mathematical tools to handle randomness and uncertainty. The usefulness of this approach will be illustrated by physical, technical and other examples.

Learning outcomes (after completion of the course the student should be able to)

After the course, the student should be well acquainted with basic probability theory and a good practice of the statistical mind-set, statistical modeling and basic statistical methods. A detailed reading display will be posted on the course homepage.

Content

Sample space, probabilities, conditional probabilities. Different probability distributions and its common applications. Means to operate on random variables, expected value, variance. Central limit theorem, law of large numbers, failure rate. Inference, maximum likelihood methods, confidence intervals, and some significance tests. Linear regression. The Poisson process.

Organisation

Lectures and exercises. Dugga that can give bonus points for the exam.

Literature

P. Olofsson and M. Andersson, Probability, Statistics and Stochastic Processes, 2nd edition, Wiley 2011.

Examination including compulsory elements

A written exam.

The course examiner may assess individual students in other ways than what is stated above if there are special reasons for doing so, for example if a student has a decision from Chalmers on educational support due to disability.